MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 7208

This specification is approved for use within the Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Pulsed, integral magnet, 16,000 to 17,200 MHz, tunable, 100 kw peak power output, air cooled, unipotential cathode.

ABSOLUTE RATINGS:

Parameter:	Ef	If (surge)	If	tk	ib	Pi	pi	epy	tpc	rrv	prr	Du
Unit:	V	a	A	sec	a	W	kw	kv	µs	kv/µs	pps	---
Maximum:	13.9	13.6	3.5	---	20	350	365	20	3.3	100	5000	0.001
Minimum:	---	---	---	270	5	---	---	---	0.20	60	---	---
	(Note 1)	(Note 2)								(Note 3)		

ABSOLUTE RATINGS:

		Temperature		Tuner	Pressurization	
Parameter:	VSWR	Input terminal	Body	stop torque	Input	Output
Unit:	---	°C	°C	in.-oz.	psia	psia
Maximum:	1.5	300	150	50	45	45
Minimum:	---	---	---	---	10	15
		(Note 4)		(Note 5)	(Note 6)	

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Mounting position:	Any
Cooling:	Forced air (see figure 2)
Mounting support:	See figure 3
Coupling:	See notes 7 and 8
Weight:	14 pounds maximum (see note 9)
Marking:	See note 10 and figure 1

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TEST CONDITION (1):

Parameter:	Ef	tk	ib	tpc	rrv	Du
Unit:	V	sec	a	µs	kv/µs	---
Maximum:	---	---	17.5	3.3	---	0.00102
	7.6	---	---	3.0	---	0.00100
Minimum:	---	270	17.0	2.7	100	0.00098
	(Note 1)			(Note 11)	(Note 3)	

FREQUENCY		
F	MHz	±
2	16,000	20
3	16,500	20
4	17,000	20

TEST CONDITION (2):

Parameter:	Ef	tk	ib	tpc	rrv	Du
Unit:	V	sec	a	µs	kv/µs	---
Maximum:	---	---	17.5	0.275	---	0.00072
	8.8	---	---	0.250	---	0.00070
Minimum:	---	270	17.0	0.225	100	0.00068
	(Note 1)			(Note 12)	(Note 3)	

TEST CONDITION (3):

Ef
V

12.6

(Note 1)

GENERAL:

Qualification - Not required

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Method	Requirement or test	Notes	Test	Conditions	Symbol	Limits		Unit
						Minimum	Maximum	
	Quality conformance inspection, part 1	13						
4289	Heater current, nonoperating		3	tk = 300 seconds	If	3.0	3.5	A
4223	Operating tuner torque	14		TA = 25°C	Torque	---	8	in.-oz.
4303	Warmup time	1	3		---	---	---	---
1367	Magnet Polarity	15			---	---	---	---
	Oscillation I		1	σ' < 1.10 except where noted				
4306	Pulse voltage	10		F = F4	epy	16	18	kv
4250	Peak power output			F = F2 to F4; t = 20 minutes to 30 minutes	po	100	---	kw
4250	Power ratio	10,16		F = F2 to F4 F = F2 to F3	ratio ratio	--- ---	1.30:1 1.25:1	--- ---
4315	Stability	10,17,18		F = F2 to F4; σ' ≥ 1.50	MP	---	1.0	%
4308	RF bandwidth	10,18		F = F2 to F4; σ' ≥ 1.50	BW	---	0.7/tpc	MHz
4308	Minor lobe ratio	10,18		F = F2 to F4; σ' ≥ 1.50	ratio	6.0	---	dB
4310	Frequency pulling figure	10,18,19		F = F2; σ' ≥ 1.50	ΔF	---	8.0	MHz
	Oscillation II	10,18	2	F = F2 to F4; σ' ≥ 1.50				
4315	Stability	17			MP	---	1.0	%
4308	RF bandwidth				BW	---	6.0/tpc	MHz
4308	Minor lobe ratio				ratio	6.0	---	dB
	Quality conformance inspection, part 2							
4003	Pressurizing	7,8		45 psia minimum, input and output	---	---	---	---
1031	Vibration	20,21		No voltages applied; t = 300 seconds				
				F = 5 to 500 to 5Hz	---	---	---	---
				F = 5 to 20 Hz; 0.25 in. double amplitude	---	---	---	---
				F = 20 to 500 Hz; 5G	---	---	---	---
	Oscillation I		1	σ' < 1.0 except where noted				
4223	Resettability	10		F = F3 ± 40 MHz	ΔF	---	3.0	MHz
4223	Tuning rate			F2 to F4; F4 to F2	Tuner turns	3.26	3.57	turns
				F2 to F3; F4 to F3	Tuner turns	1.70	2.07	turns
4030	Thermal frequency drift	10,22		F = F4; t = 10 minutes	ΔF	---	15	MHz
4306	Pulse voltage	10		F = F2 to F4	Δepy	---	0.8	kv

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Method	Requirement or test	Notes	Test	Conditions	Symbol	Limits Minimum	Limits Maximum	Unit
	Quality conformance inspection, part 2 - Oscillation I - Continued							
4315	Starting stability	10,17 23,24		F = F3 ± 40 MHz; σ' = 1.50; Ef = 11.9V	MP	---	1.5	%
	Oscillation II		2	σ' < 1.10				
4250	Peak power output			F = F2 to F4; t = 20 minutes to 30 minutes	po	100	---	kw
4250	Power ratio	10,16		F = F2 to F4 F = F2 to F3	ratio ratio	--- ---	1.30:1 1.25:1	--- ---
	Quality conformance inspection, part 3							
4551	Intermittent life	25-28		Group D	t	500	---	hrs
---	Intermittent life-test end points:		1	F = F2 to F4				
4250	Peak power output			t = 20 minutes to 30 minutes; σ' < 1.10	po	80	---	kw
4308	RF bandwidth	10,18		σ' ≥ 1.50	BW	---	1.0/tpc	MHz
4315	Stability	10,17,18		σ' ≥ 1.50	MP	---	2.0	%
---	Intermittent life-test end points:		2	F = F2 to F4; σ' ≥ 1.50				
4308	RF bandwidth	10,18			BW	---	8.0/tpc	MHz
4315	Stability	10,17,18			MP	---	2.0	%
4027	Temperature coefficient	4,10, 29-31	1	F = F3; σ' < 1.10; body T = 75° to 125°C	ΔF/ΔT	---	-0.5	MHz/°C
4223	Mechanical tuning fatigue	29,32,33	3	Duration of each cycle = 1 to 10 seconds	---	5000	---	cycles
4223	Tuner stop endurance	29, 34			---	---	---	---
1031	Vibration	10,20,29	1	σ' < 1.10; F = F2 to F4; t = 300 seconds F = 5 to 500 to 5 Hz F = 5 to 20 Hz; 0.25 in. double amplitude F = 20 to 500 Hz; 5 G	ΔF	---	3.0	MHz
1042	Shock, specified pulse	29, 35		Half sine; 50G; 4 ms; 0.25 amplitude	---	---	---	---
1143	Forced cooling	4,29,31		TA = 50°C max.; air flow = 10 cfm; P_p = Pi - Po = 200W	ΔT(body)	---	100	°C
4223	Starting tuner torque	29	3	TA = -55°C; tk = 300 seconds min.	Torque	---	25	in.-oz.
---	Atmosphere frequency shift	10,29,36		No voltages	ΔF	---	9	MHz

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NOTES:

1. Prior to the application of high voltage, the cathode shall be heated by applying 13.2 V for 270 seconds minimum for test condition 1 and 12.6 V for 270 seconds minimum for test conditions 2 and 3. On application of anode voltage, the heater voltage shall be reduced according to the following schedule:

Ef	Ib
12.6 V	0 mA
10.5 V	5 mA
9.3 V	10 mA
8.1 V	15 mA
6.4 V	20 mA

2. There shall be a suitable series impedance during starting time to limit surge current to this value. The cold resistance of the heater is approximately 0.5 ohm.

3. The rrv is defined by the steepest tangent to the leading edge of the voltage pulse above 50 percent amplitude. Any capacitance used in the viewing system shall not exceed 6.0 pF.

4. Temperature shall be measured at the point shown on figure 2.

5. The tuning mechanism shall be capable of withstanding a static torque of 50 in.-oz at the end of its travel.

6. The TUT shall be capable of normal operation without electrical breakdown, with the input bushing in air at normal atmospheric conditions.

7. With the output flange clamped against the modified choke flange (UG-541/U with clearance holes instead of tapped holes), using the gasket (Synthetic rubber part no. MS90064-12 or equivalent), and with the specified air pressure applied to the interior of the waveguide, the internal-to-external leak rate shall not exceed 0.1 cubic inch per minute.

8. With the base plate clamped against the mounting plate (see figure 3), using gasket MS-28775-233, and with the specified air pressure applied so as to surround the entire input terminal beyond the base plate, the internal-to-external leak rate shall not exceed 0.1 cubic inch per minute.

9. The tube shall be weighed with unmagnetized magnets, and without container, packing materials, dust covers, or accessories.

10. In addition to the regular markings, code letters and dial settings shall be marked on the tube to indicate positions of the tuner drive shaft for the following frequencies under Oscillation I and with the body temperature at 40°C ± 10°C. The tuner shall be rotated in a clockwise direction for each reading. See notes 4 and 31.

Frequency ±20 MHz	Code Letter
16,000 MHz	F2
16,500 MHz	F3
17,000 MHz	F4

11. The current amplitude at the leading edge of the pulse shall be not less than the current amplitude at the trailing edge of the pulse, disregarding the spike and ripple. The current pulse shall satisfy the following requirements at F3:

trc < 0.140 µs	ripple < ±20%
tfc < 0.40 µs	droop < 3a

12. The current amplitude at the leading edge of the pulse shall be not less than the current amplitude at the trailing edge of the pulse, disregarding both the spike and ripple. The current pulse shall satisfy the following requirements at F3:

trc < 0.040 µs	ripple < ± 10%
tfc < 0.120 µs	droop < 2a

13. The AQL (percent defective) for all tests in QCI, part 1, shall be 1.0, inspection level S3.

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NOTES: - Continued

14. This test shall be performed after subjection to mechanical fatigue. The tuner drive mechanism shall not be "set" against either mechanical stop. The measurement shall then be made at any speed between 20 and 30 rpm.

15. The tube shall be tested to determine that the south-seeking pole is adjacent to the cathode terminal.

16. The ratio of maximum to minimum power output across the stated frequency bands shall not exceed the specified limits for constant anode current.

17. The missing pulses, due to any cause, are considered to be "missing" if the rf energy is less than 75 percent of the normal energy level in a ±1% frequency range of the normal operating frequency. The VSWR shall be adjusted to that phase producing maximum instability and the missing pulses shall be counted during the last 3 minutes of a 6 minute test period.

18. The input bushing temperature shall be maintained between 240°C and 300°C. The temperature shall be measured at the point indicated on figure 2.

19. Requirements shall be met over the tuning range of F2 through F4.

20. The TUT shall be fastened rigidly to the vibration platform and vibrated with simple harmonic motion at frequencies and G's or displacements as specified. The direction of vibration shall be in three mutually perpendicular planes, one of which shall be parallel to the reference plane of the base plate. The change in frequency with respect to time shall approximate a logarithmic curve. After the tests, the TUT shall show no mechanical failure and shall meet the requirements of Oscillation I of QCI, part 1.

21. On evidence of satisfactory quality, the inspector may limit this test to ten tubes per month if production is continuous. Normal design testing shall be resumed if defective tubes exceed 6 percent of production.

22. Ten cubic feet per minute of ambient temperature air shall be directed through the cooling fins by means of a fixture as per figure 2 or equivalent. The frequency shall be measured immediately after the application of anode voltage. The frequency change in a positive direction shall not exceed the specified amount during the ten minute period thereafter.

23. This test shall be performed immediately after the nonoperation shelf period. QCI, part 1, Oscillation I stability test shall be made immediately thereafter.

24. The anode voltage shall be applied 270 seconds after the application of Ef. Ef shall then be reduced to 6.4 V.

25. This test shall be conducted in accordance with the following cycle:

Condition	Duration	Ef	Du	ib
Standby	5 minutes	12.6 V	---	0
Oscillation II	3 minutes	8.8 V	.0007	17a
Oscillation I	22 minutes	7.6 V	.001	17a
Off	15 minutes, min.	0.0 V	---	0

26. The frequency shall be changed from F2 to F4, 200 MHz every 8 radiate hours, either continuously or in 200 MHz increments.

27. At least one of the tubes in the monthly life test sample shall be tested every other month, to the end of life.

28. $\sigma' \geq 1.50$ phase shall vary through a minimum of 180 electrical degrees every 30 minutes of high voltage time.

29. This test shall be performed during the initial production and once each succeeding 12-calendar month period in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes having an acceptance number of zero, and a second sample of three tubes having a combined acceptance number of one. In the event of failure, the test shall be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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NOTES: - Continued.

30. The TUT shall be placed in a suitable enclosed container to obtain thermal equilibrium.

31. Temperature measurements shall be made only after thermal equilibrium has been reached.

32. Separate samples shall be required for this test.

33. The TUT shall meet the requirements of the QCI, part 1, Oscillation I tests, and the resettability and operating tuner torque tests at the completion of this test.

34. Apply between 50 and 55 in.-oz. of torque to the tuner drive shaft. The radial and axial force applied to the tuner drive shaft shall not exceed one pound during this test.

35. This test shall be performed on the High Impact Shock Machine for Electronic Devices (see drawing 180 JAN) or equivalent as follows:

    a. A resilient cushion of suitable size and hardness shall be interposed between hammer and anvil of table and a suitable hammer angle selected to produce a shock of the specified magnitude and duration. The mounting plate of the TUT shall be bolted with nonmagnetic metallic bolts to either the table or the standard angle bracket, depending on the direction of the desired shock, using a 1-9/16 in. (39.68 mm) nonmagnetic metallic spacer between the tube mounting plate and the table or angle bracket. The shock shall be measured on the nonmagnetic spacer. The tube shall be given one shock in each of the following directions:

        (1) Parallel to the cathode axis, with the cathode terminals pointing away from the hammer.

        (2) Perpendicular to the catnode and output waveguide axes.

        (3) Perpendicular to the cathode axis and parallel to the output waveguide axis.

    b. After this test the TUT shall show no mechanical failure and shall meet the requirements of the QCI, part 1, Oscillation I tests.

36. The resonant frequency of the TUT shall be measured with a local oscillator and frequency measuring device with the tuner dial set on the TUT's frequency code letter F4. The atmospheric pressure on the tuning mechanism shall then be reduced to 0.55 psia maximum, and the resonant frequency remeasured.

37. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - ER

Review activities:
Army - AR
DLA - ES

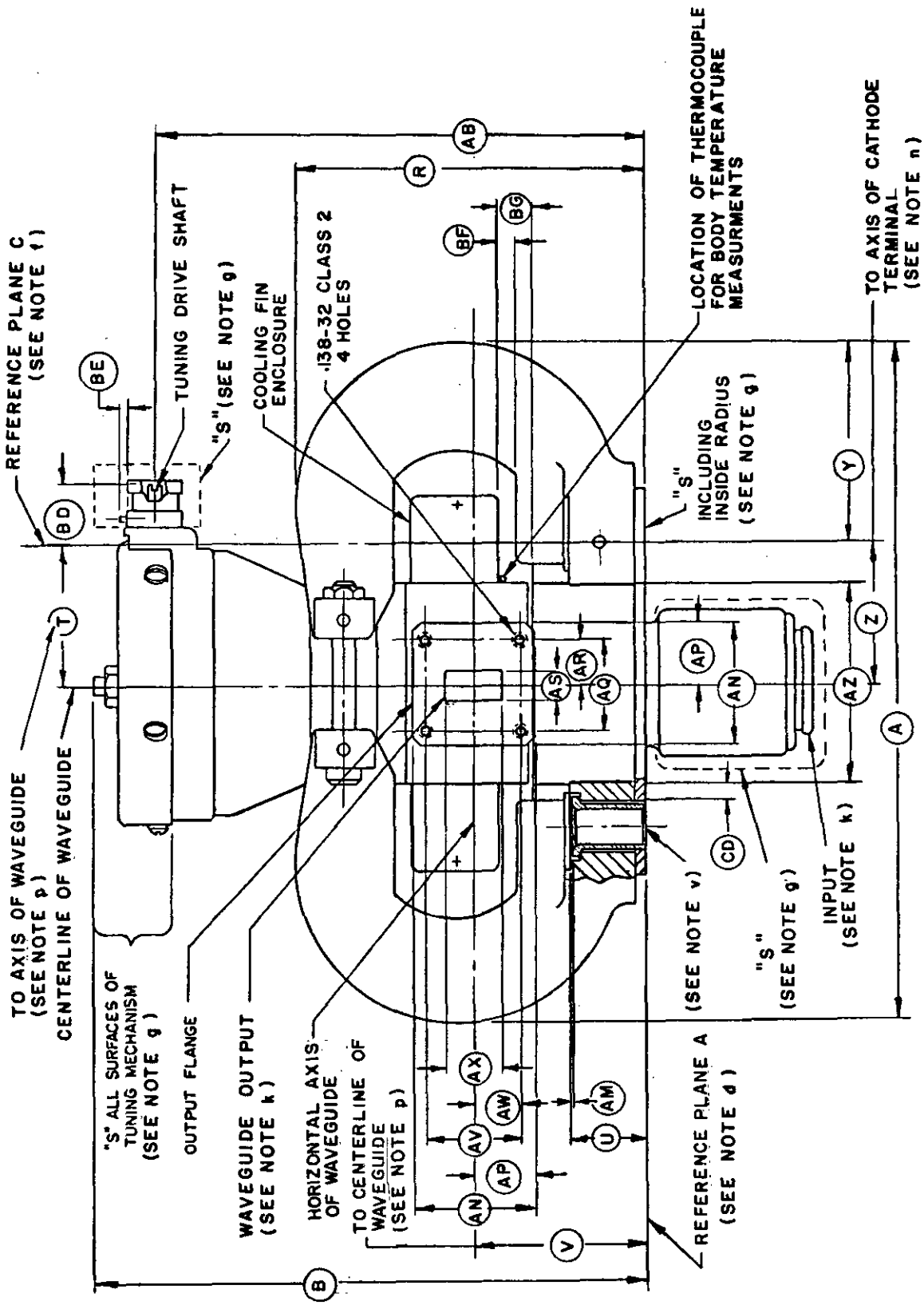
User activity:
Army - MI

Preparing activity:
Army - ER

Agent:
DLA - ES

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FRONT VIEW

FIGURE 1. Outline drawing of electron tube type 7208.

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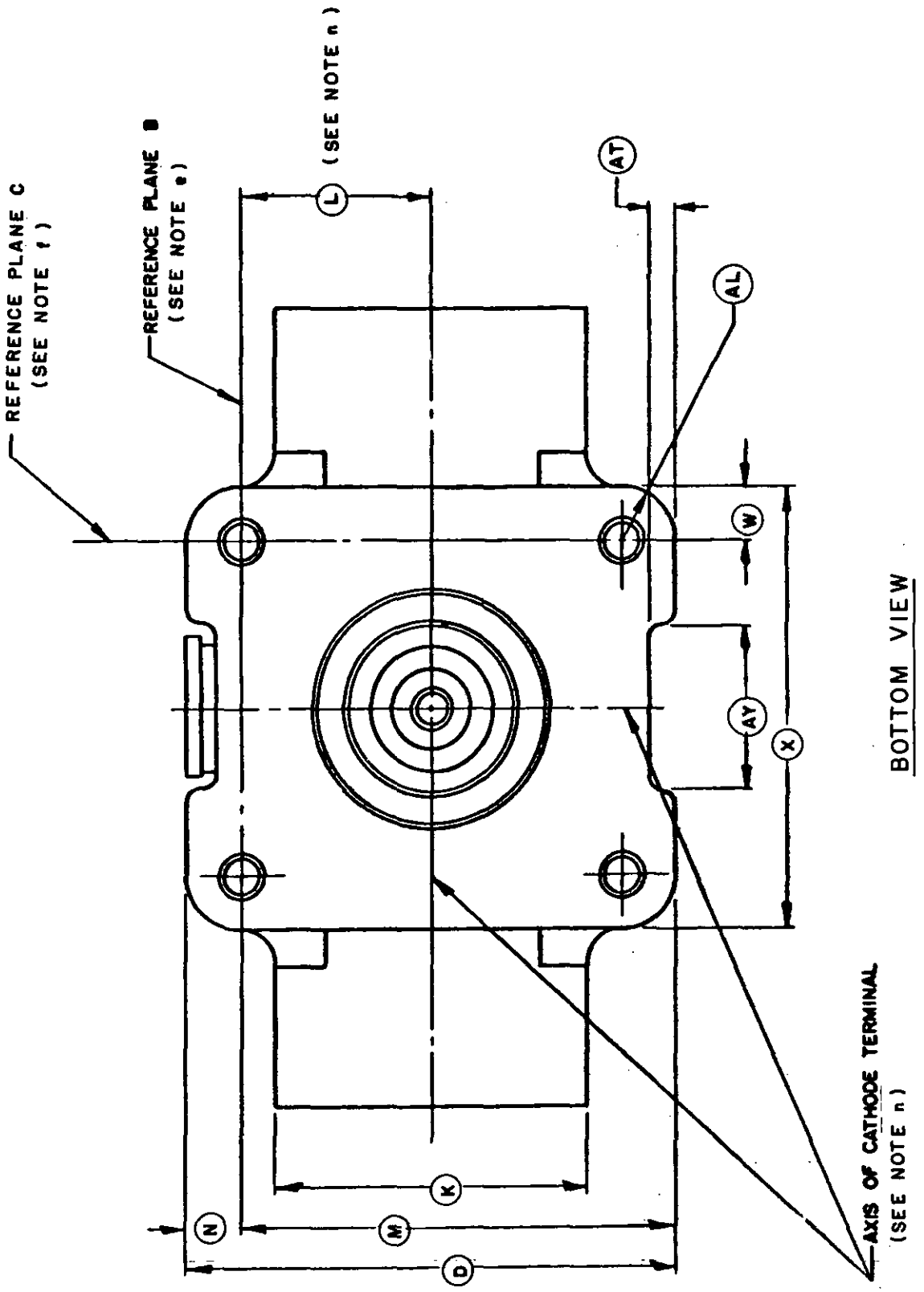


FIGURE 1. Outline drawing of electron tube type 7208 - Continued.

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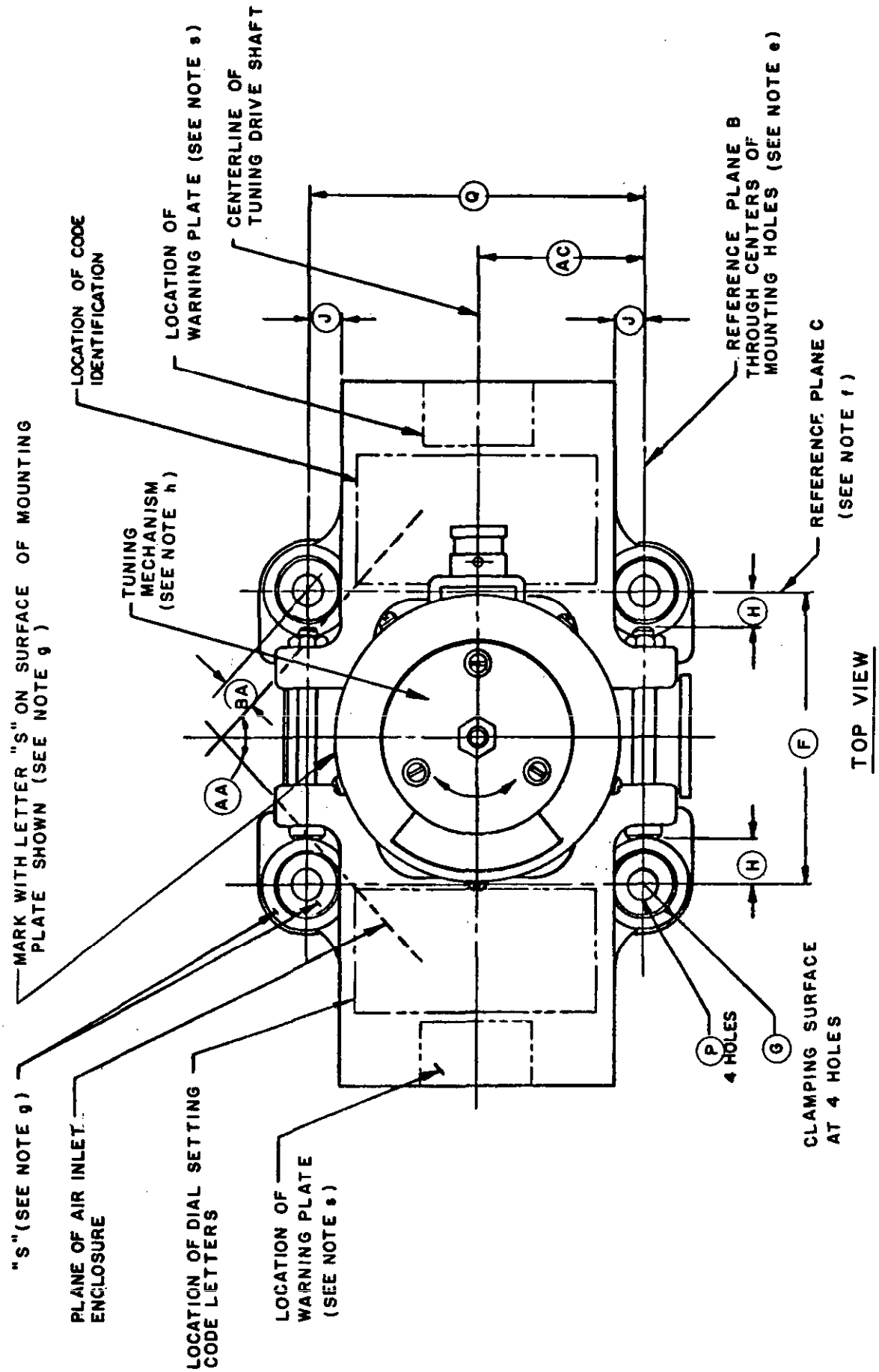


FIGURE 1. Outline drawing of electron tube type 7208 - Continued.

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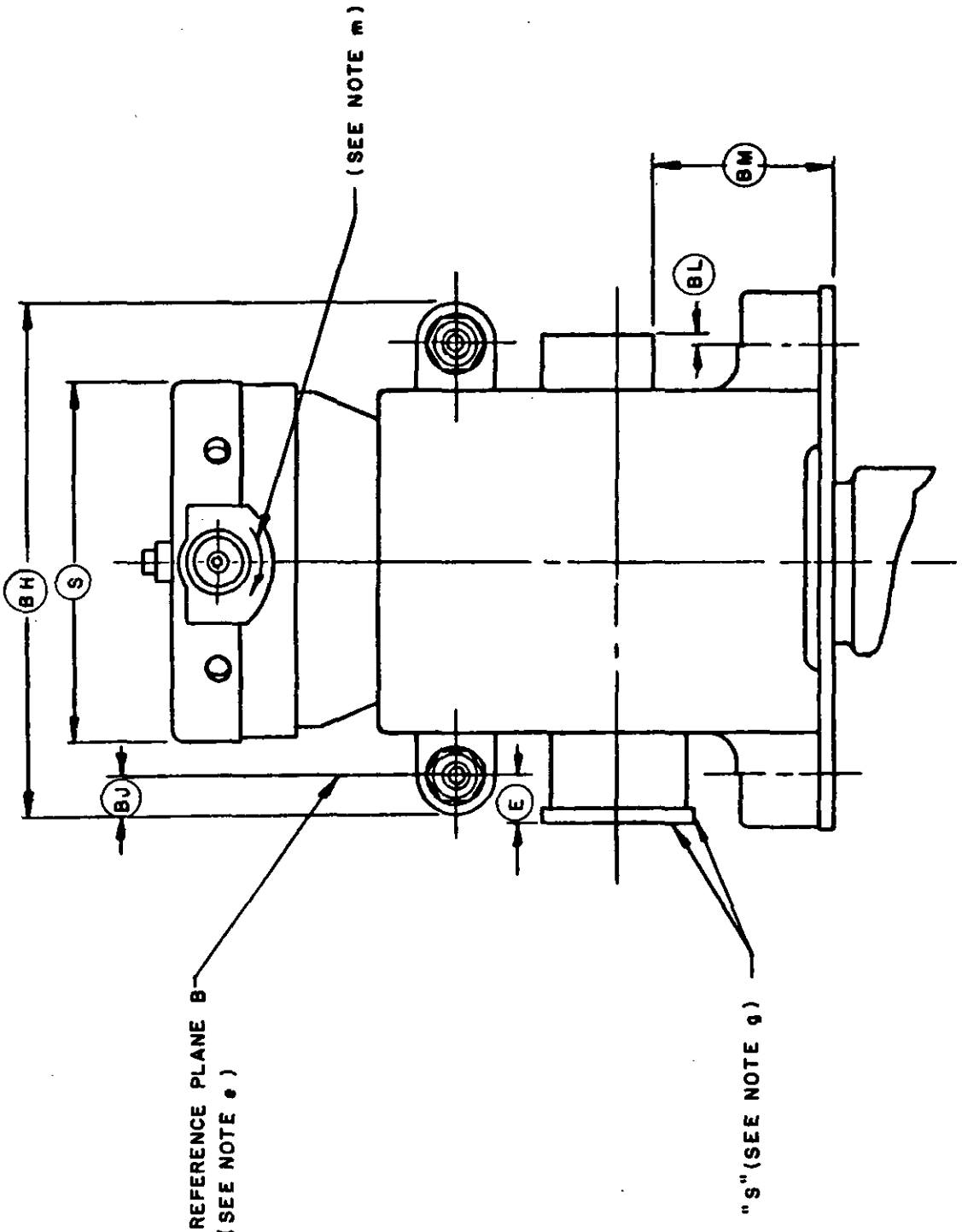
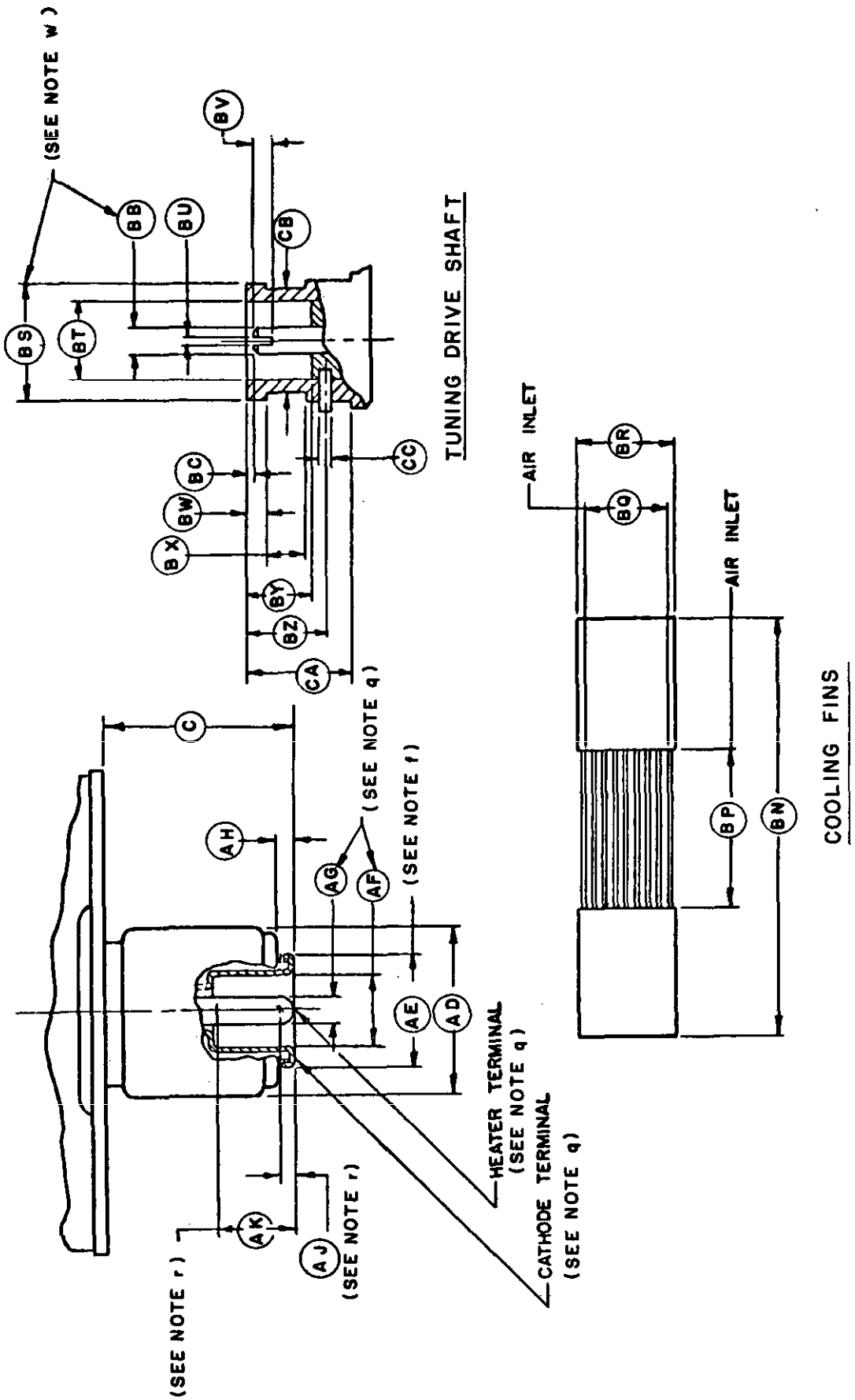


FIGURE 1. Outline drawing of electron tube type 7208 - Continued.

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DETAIL OF COOLING FINS AND TUNING DRIVE SHAFT

FIGURE 1. Outline drawing of electron tube type 7208 - Continued.

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Ltr	Dimensions are in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
E	.305 (7.75)	.365 (9.27)
F	2.990 (75.95)	3.010 (76.45)
G	.468 (11.89) rad	— — —
H	.312 (7.92)	— — —
L	1.750 (44.45) (see note n)	
P	.339 (8.61)	.349 (8.86)
Q	3.490 (88.65)	3.510 (89.15)
U	.844 (21.44)	.906 (23.01)
V	1.781 (45.24)	1.843 (46.81)
AB	5.188 (131.78)	5.312 (134.92)
AC	1.719 (43.66)	1.781 (45.24)
AD	— — —	1.625 (41.28) dia
AE	1.047 (26.59) dia	1.098 (27.89) dia
AM	— — —	.015 (0.38)
AP	.625 (15.88)	.687 (17.45)
AQ	.952 (24.18)	.960 (24.38)
AR	.476 (12.09)	.480 (12.19)
AS	.309 (7.85)	.313 (7.95)
AV	.991 (25.17)	.999 (25.37)
AW	.4955 (12.586)	.4995 (12.687)
AX	.620 (15.75)	.624 (15.85)
BA	.375 (9.52)	— — —
BB	.124 (3.15) dia	.125 (3.18) dia
BC	.021 (0.53)	.041 (1.04)
BD	.612 (15.54)	.672 (17.07)
BE	.047 (1.19)	.093 (2.36)
BM	1.484 (37.69)	1.609 (40.87)
BP	1.406 (35.71)	1.530 (38.86)
BR	.906 (23.01)	.968 (24.59)
BS	.560 (14.22) dia	.564 (14.33) dia
BT	.370 (9.40) dia	.380 (9.65) dia
BU	.041 (1.04)	.045 (1.14)
BV	.084 (2.13)	.104 (2.64)
BW	.089 (2.26)	.099 (2.51)
BX	.183 (4.65)	.193 (4.90)
BY	.313 (7.95)	.343 (8.71)
BZ	.360 (9.14)	.390 (9.91)
CA	.490 (12.45)	.510 (12.95)
CC	— — —	.094 (2.39)

Ltr	Dimensions are in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
C	1.813 (46.05)	1.937 (49.20)
M	— — —	4.031 (102.39)
N	— — —	.531 (13.49)
T	1.469 (37.31)	1.531 (38.89)
W	.469 (11.91)	.531 (13.49)
Y	— — —	2.250 (57.15)
Z	1.500 (38.10) (see note n)	
BF	.125 (3.18)	— — —
BG	.265 (6.73)	— — —
BH	— — —	4.375 (111.12)
BJ	— — —	.437 (11.10)
BL	.031 (0.79)	.155 (3.94)
CD	.093 (2.36)	— — —
Periodic check (see note u)		
A	— — —	7.500 (190.50)
B	— — —	6.000 (152.40)
D	— — —	4.562 (115.87)
J	.250 (6.35)	— — —
K	— — —	3.000 (76.20)
R	— — —	4.000 (101.60)
S	2.969 (75.41) dia	3.250 (82.55) dia
X	— — —	4.062 (103.17)
AF	.747 (18.97) dia	.753 (19.13) dia
AG	.247 (6.27) dia	.253 (6.43) dia
AH	.125 (3.18)	— — —
AJ	— — —	.156 (3.96)
AK	.750 (19.05)	— — —
AN	1.281 (32.54)	1.343 (34.11)
AY	1.469 (37.31)	1.531 (38.89)
Nominal dimensions (see note b)		
AA	84°	
AL	.500 (12.70) rad	
AT	.250 (6.35)	
AZ	2.125 (53.98)	
BN	3.937 (100.00)	
BQ	.812 (20.62)	
CB	.500 (12.70)	

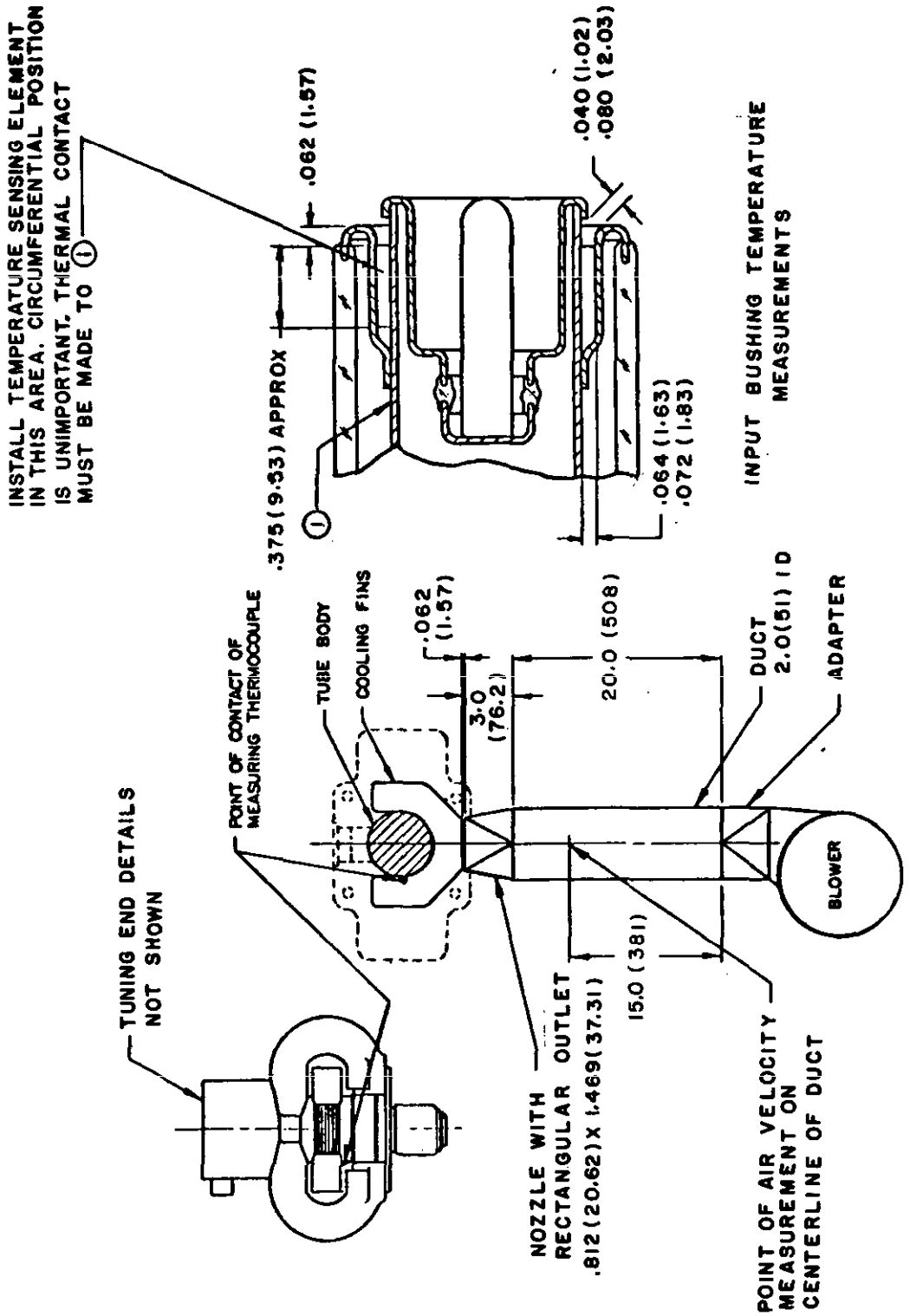
FIGURE 1. Outline drawing of electron tube type 7208 - Continued.

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NOTES:
a. The protective cover over the tuning head must be removed before the tube is used.
b. Dimensions without tolerance are for information only and are not required for inspection purposes.
c. The tube is provided with an internal stop for a maximum of 50 impacts caused by unrestrained tuner drive motion.
d. Reference place A is defined as that plane which is coincident with the face of the mounting surface as shown.
e. Reference plane B is defined as that plane which is perpendicular to reference plane A, and which passes through centers of mounting holes as shown.
f. Reference plane C is defined as that plane which is mutually perpendicular to reference planes A and B, and which passes through centers of mounting holes as shown.
g. All metal parts except those designated "S" shall be protected with paint or other suitable finish.
h. Gear and pinion of the tuning mechanism shall be free from corrosion, paint, or other obstructions. The tuning mechanism shall rest against the stop above the F4 frequency for shipment.
j. The inner laminations of the cooling fins are not painted, but a slight overspray of the enamel is allowed.
k. The opening of the waveguide input and output shall be enclosed by dust covers when the tube is not in use.
m. The frequency shall increase when the drive shaft is driven in the direction indicated by the arrow.
n. The axis of the cathode terminal shall occupy the specifed location within a 0.047 inch (1.19 mm) radius. Limits include angular and lateral deviations.
p. Limits include angular and lateral deviations.
q. Heater and cathode terminals shall be concentric within 0.010 inch (0.25 mm).
r. These dimensions define the extremities of the cylindrical sections given by the "AG" and "AF" dimensions.
s. Warning: Maintain minimum clearance of 4 inches between magnet and magnetic materials (magnets, steel tools, plates, etc.). Use no magnetic inspection tools.
t. There shall be no clamping on this diameter.
u. These dimensions shall be checked during the initial production and once each succeeding 12-calendar month period in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes having an acceptance number of zero, and a second sample of three tubes having a combined acceptance number of one. In the event of failure, the dimensions shall be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.
v. Hollow bolt flush or underflush with respect to both the reference plane and clamping surface.
w. These diameters shall be concentric within 0.055 inch (1.40 mm) TIR.

FIGURE 1. Outline drawing of electron tube type 7208 - Continued.

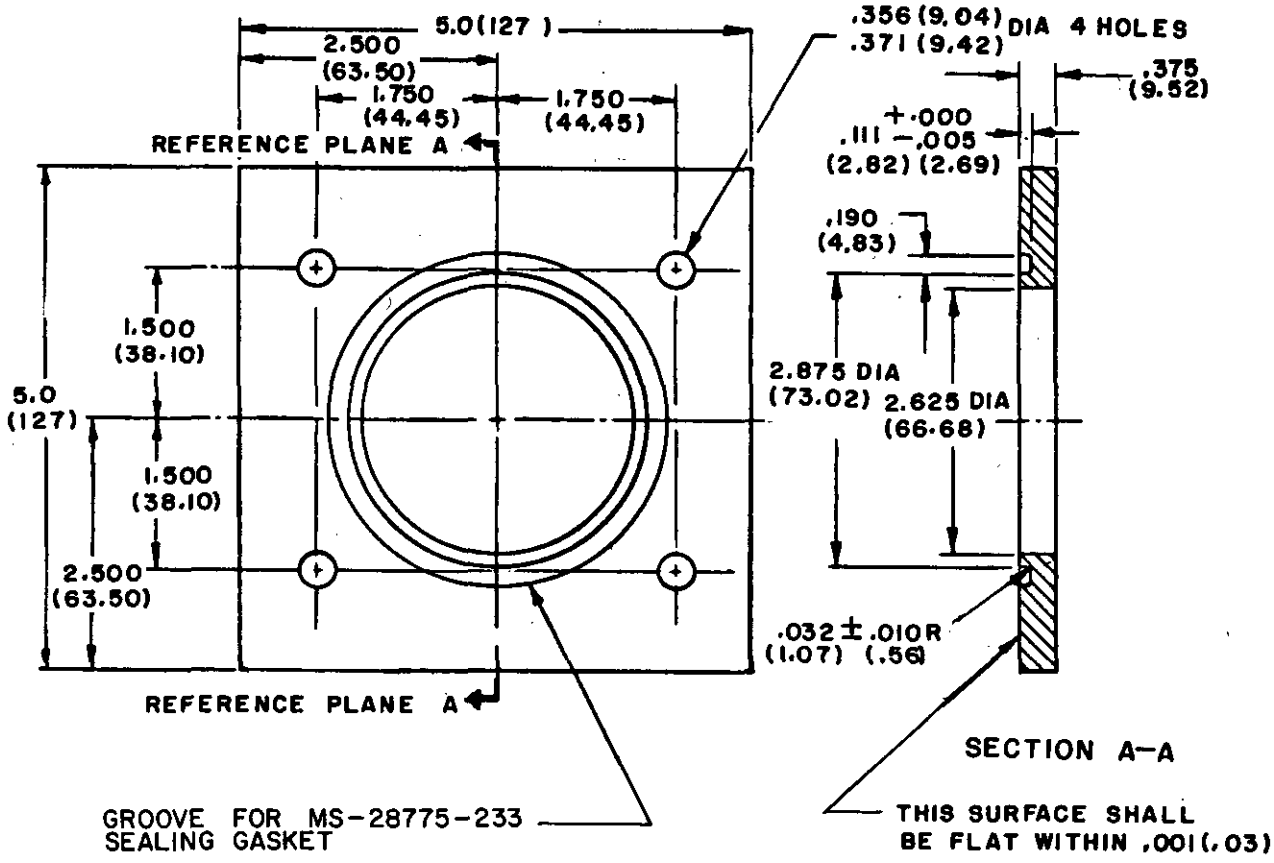
Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:03Z
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NOTES:
a. Dimensions are in inches.
b. Metric equivalents are in parentheses.
c. The dimensions listed on the input bushing temperature section are for information only and shall be used for designing suitable devices for temperature.

FIGURE 2. Cooling arrangement.

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NOTES:
a. Dimensions are in inches.
b. Metric equivalents are in parentheses.

FIGURE 3. Mounting plate.

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